

Zechuan Shi

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SUMMARY

Computational Biologist with 6+ years of experience specializing in AI-augmented multi-omic analysis and large-scale infrastructure development. Develops network-based and machine learning models to characterize genetic perturbations and robust analytical pipelines for single-cell multi-omics. Architected workflows for 2.3M+ nuclei datasets and contributed to high-impact open-source software like *hdWGCNA*, *scROAD* and *compact*. Authored 15+ peer-reviewed publications in neurodegenerative disorders, including *Science Advances*, *Nature Genetics*, and *Nature Neuroscience*. Highly proficient in Python, R, HPC, and cloud-based computing to drive functional genomics discovery, systematic target prioritization, and large-scale perturbation analysis.

EDUCATION

Ph.D. Candidate in Bioinformatics, University of California, Irvine Jun 2026

Research Direction: Computational modeling, Machine Learning, Genomics, GWAS, Single Cell RNA/ATAC/Spatial

Master of Science in Biotechnology, Johns Hopkins University, GPA: 3.92/4.0 May 2018

Concentration: Bioinformatics, Molecular Targets and Drug Discovery Technologies, Oncology

TECHNICAL SKILLS

Genomics & Transcriptomic Profiling: Single-cell RNA-seq, bulk RNA-seq, spatial transcriptomics, Xenium, ATAC-seq, proteomics, flow cytometry data analysis, multimodal data integration, large-scale clinical and NGS datasets

Programming & Computational Infrastructure: R, Python, Bash, Nextflow, SQL, Git version control and collaboration, method and pipeline optimization, reproducible workflows, pipeline development, code documentation

Computational Genomics & Modeling: Differential gene expression analysis, on-target and off-target effect characterization, gene regulatory network modeling, perturbational genomics, Perturb-seq analysis, causal inference, GWAS, eQTL analysis, biomarker discovery, survival analysis

Machine Learning & AI: Machine learning and AI-augmented analysis workflows, transformer-based models, variational autoencoder models, diffusion models, graph-based modeling for gene regulatory networks, LLM-assisted development, foundation models for genomics, Geneformer, scGPT, STATE

Therapeutic & Disease Domain Knowledge: Gene therapy and RNA-based perturbations, cancer biology, immunology, neurodegenerative disease

Software Development:

- **Compact** (Lead Developer): an in-silico module perturbation method that unlocks a functional understanding of dynamic gene networks in single-cell data (<https://smorabit.github.io/compact/>)
- **scROAD** (Lead Developer): a database portal that offers single-cell cCREs' transcription factor occupancy information generated from snATAC-seq analysis of human postmortem prefrontal cortex (PFC) tissue for dementia research (<https://swaruplab.bio.uci.edu/scROAD/>)
- **ArchRtoSignac** (Lead Developer): an R package that allows easier implementation of both ArchR and Signac scATAC-seq analysis pipelines (<https://github.com/swaruplabUCI/ArchRtoSignac>)
- **hdWGCNA** (Contributor): an R package for performing weighted gene co-expression network analysis (WGCNA) in high dimensional transcriptomics data (470+ stars |<https://github.com/smorabit/hdWGCNA>)

SELECTED PUBLICATIONS (15+)

- **Shi, Z.***, Morabito, S.*, et al. (2026). In silico module perturbation analysis unlocks a functional understanding of the dynamic gene networks in single-cell data. (*bioRxiv*)
- **Shi, Z.**, Das, S., Morabito, et al. (2025). Single-nucleus multi-omics identifies shared and distinct pathways in Pick's and Alzheimer's disease. *Science Advances*.
- Fotio, Y., Al Masri, S., **Shi, Z.**, et al. (2026). Metabolic reallocation in spinal cord oligodendrocytes drives chronic pain via neuronal β -amyloid production. (Manuscript in review; *Science Translational Medicine*)

- Tran, K.M., Kwang, N.E., Butler, C.A., [et al., including **Shi, Z.**], LaFerla, F., ... Green, K.. (2025). APOE Christchurch enhances a disease-associated microglial response to plaque but suppresses response to tau pathology. **Molecular Neurodegeneration**.
- Miyoshi, E., Morabito, S., Henningfield, C.M. [et al., including **Shi, Z.**]. (2024). Spatial and single-nucleus transcriptomic analysis of genetic and sporadic forms of Alzheimer's Disease. **Nature Genetics**.
- Tiwari, V., Prajapati, B., Asare, Y. [et al., including **Shi, Z.**]. (2024). Innate immune training restores pro-reparative myeloid functions for remyelination in the aged central nervous system. **Immunity**, 57(9), 2173–2190.e8.
- O'Shea, T.M., Ao, Y., Wang, S. [et al., including **Shi, Z.**]. (2024). Derivation and transcriptional reprogramming of border-forming wound repair astrocytes after spinal cord injury or stroke in mice. **Nature Neuroscience**.
- **Shi, Z.***, Das, S.*, Morabito, S., Miyoshi, E., & Swarup, V. (2022). Protocol for single-nucleus ATAC sequencing and bioinformatic analysis in frozen human brain tissue. **STAR Protocols**, 3(3), 101491.

COMPUTATIONAL EXPERIENCE

Graduate Student Researcher, University of California – Irvine, CA

Sep 2020 – Jun 2026

Focus: Computational genomics and multi-omics modeling for therapeutic target discovery and perturbation analysis

- Applied transformer and VAE frameworks to model gene regulatory structure and perturbation responses, and developed **Compact**, a network-based approach to prioritize therapeutic targets by modeling on-target efficacy and transcriptome-wide effects across oncology and neurodegenerative datasets.
- Integrated snRNA-seq, snATAC-seq, and spatial transcriptomics across 2M+ cells from human disease cohorts to characterize cell-type-specific gene expression changes and identify molecular signatures associated with disease progression.
- Leveraged AI-augmented workflows and genomic foundation models to accelerate transcriptomic analysis, enabling systematic exploration of therapeutic effects and improving analytical efficiency.
- Developed reproducible computational pipelines, internal tools, and analysis workflows for large-scale NGS and multi-omics data, improving analytical efficiency, standardization, and documentation across projects.
- Collaborated with experimental and clinical teams to translate computational analyses into actionable hypotheses for therapeutic development, contributing to 15+ peer-reviewed publications.

PROFESSIONAL EXPERIENCE

Biotech Investment Fellow, BlueRun Ventures – Beijing, China (Remote)

Nov 2021 – Jan 2022

- Evaluated early-stage biotech companies in gene therapy, drug discovery, and computational biology, assessing technology feasibility, therapeutic potential, and development risks.
- Translated complex scientific innovations into actionable insights for investment decisions, analyzing competitive landscapes and communicating findings to cross-functional stakeholders.

Preclinical Research Specialist, University of Pennsylvania – Philadelphia, PA

Jul 2018 – Jul 2020

Pharma Academic Partnership for Recurrent Breast Cancer Drug Development

- Served as a computational liaison between academic and pharmaceutical R&D teams, translating genomic analyses into actionable insights for therapeutic development. Presented results to medicinal chemistry and clinical teams to support target prioritization and drug evaluation.
- Designed and executed large-scale preclinical studies 600+ mice, 11 HER2+ transgenic lines, and led bulk RNA-seq analysis to characterize treatment response, identify resistance mechanisms, and evaluate therapeutic efficacy.
- Characterized tumor-infiltrating immune cells and circulating tumor cells using integrated flow cytometry and transcriptomics analysis
- analyzed transcriptomic responses to siRNA perturbations in cell culture to assess target engagement and unintended gene expression changes, informing biomarker development and treatment response prediction.

SELECTED PRESENTATIONS

- “The new era of bioinformatics: How foundation models will rewrite the game in functional genomics”. **NeuroBlitz**, Irvine, CA. Nov 2025. (Talk)
- “Recent advances in Perturb-seq”. Single Cell Journal Club at **UCLA, Daniel Geschwind Lab**, Los Angeles, CA. Oct 2025. (Invited Talk)

- “In silico network reprogramming of oligodendrocytes reveals therapeutic potential in Alzheimer’s disease”. **Annual Meeting of American Society of Human Genetics**, Boston MA. Oct 2025. (**Reviewers’ Choice Abstract Award**)
- “In Silico Module Perturbation Analysis unlocks a functional understanding of the dynamic gene networks in single-cell data”. **Annual Meeting of American Society of Human Genetics**, Denver, CO. Nov 2024. (Talk)
- “Single-nucleus open chromatin accessibility landscape of Pick’s and Alzheimer’s disease”. **Alzheimer’s Association International Conference**, San Diego, CA. Jul-Aug 2022. (Talk)

FELLOWSHIP & AWARDS

James L. McGaugh Excellence Award (2026): Excellence in Research Award

ASHG Reviewers’ Choice Abstract Award (2025): Awarded to the top 10% of submissions as scored by reviewers

Biological Sciences Travel Award (2023, 2024, 2025): University of California Conference Travel Award (\$900)

Bio2X Outstanding Trainee Award (2021-2022): BlueRun Ventures Bio2X Life Sciences Ideation Award

Cum Laude (2015): Graduating with Honor, TSU

Gerhardt Research Fellow (2015): a competitive research award (\$3,000)

Grants-in-Aid of Scholarship and Research (GIASR) (2014): a competitive research grant for the initial stage of an individual student research project (\$750)

President’s Honorary Scholarship (2011-2015): an annual academic achievement scholarship, TSU (\$18,000)

TEACHING & LEADERSHIP EXPERIENCE

Program Manager, genPALs and UCI GPS – STEM Data Science Cohort – Irvine, CA Jul 2022 – Jun 2024

- Organized bi-weekly genomics and data science seminars, facilitating engagement between academic researchers and industry professionals.
- Developed and taught workshops on introductory machine learning and statistical methods for single-cell transcriptomic analysis.

Graduate Teaching Assistant, University of California – Irvine, CA Mar 2022 – Mar 2024

Classes: Bio 38: Mind, Memory, and Brain; Bio 48: The Mind-Body Connection in the Neuroscience of Well-Being; N 138: Sex Influences on Brain; NEURBIO 227. Bioinformatics and Systems Biology

- Graded 200+ students’ writing assignments, programming tasks and exams
- Facilitated in-class discussion sections and answered students’ questions